ANT 6973

# DATA VISUALIZATION AND EXPLORATION

WEEK1

# INTRODUCTION, SETUP, AND R BASICS

# PLAN FOR TODAY

1. Introductions
2. Orientation
    - Syllabus and course website
    - R and RStudio
3. A few R basics
4. Complete your first in-class activity!

# WHO AM I?

**Dr. Fernando Campos**

- Department of Anthropology
- Email: fernando.campos@utsa.edu
- Office Hours:
  - Wed, 12:30 pm – 1:30 pm
  - By appointment



# WHAT DO I DO?

## Dr. Fernando Campos

I’m a biological anthropologist, studying the behavioral ecology of wild primates.



# WHY AM I TEACHING THIS COURSE?

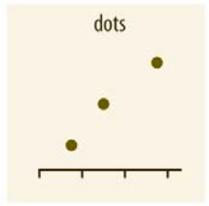
# DIRECTORY OF VISUALIZATIONS

## Amounts





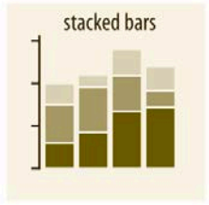








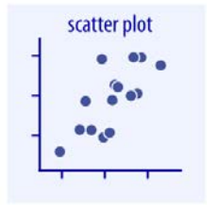


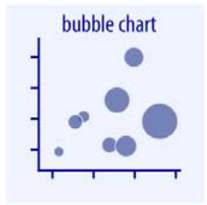







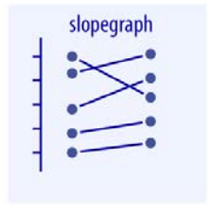


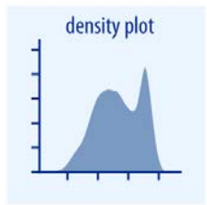


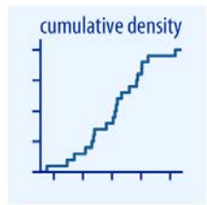





## Distributions

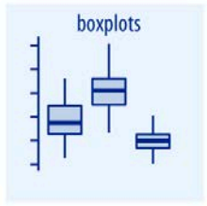


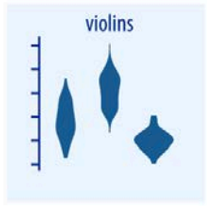



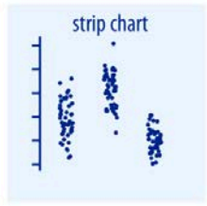









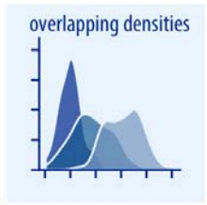













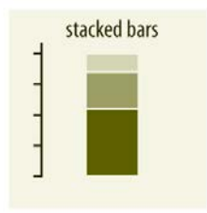
## Proportions







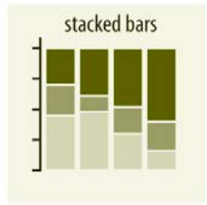


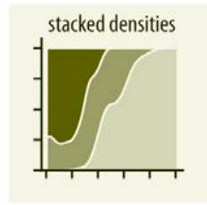











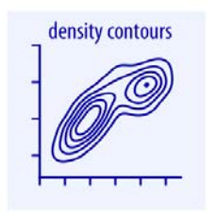


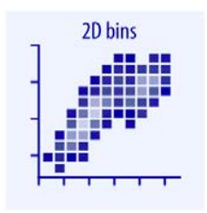


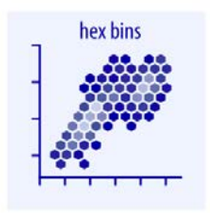


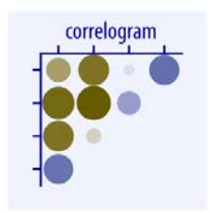


## X-Y Relationships



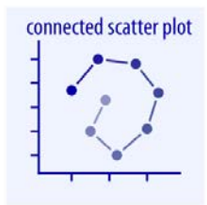



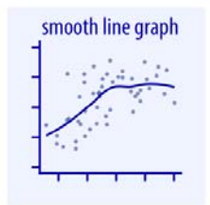



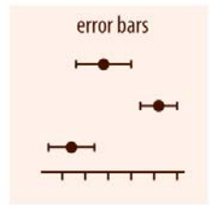






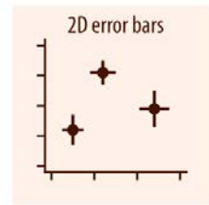



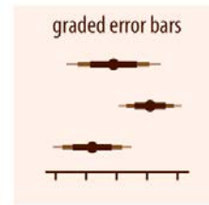

## Uncertainty





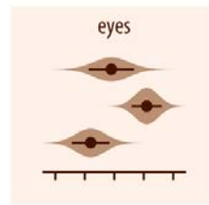


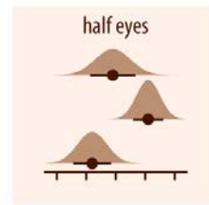


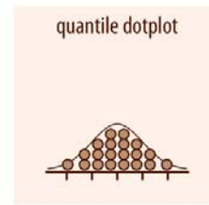









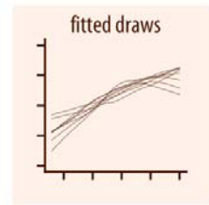










Claus Wilke, *Data Visualization*, Ch. 5: "Directory of Visualizations"

# DATA LITERACY



# YOUR TURN

1. Who are you?
2. What are your research interests?
3. What is your experience with data?
4. Do you have any experience using R, and if so, how much?
5. Is there anything specific that you hope to learn in this course?



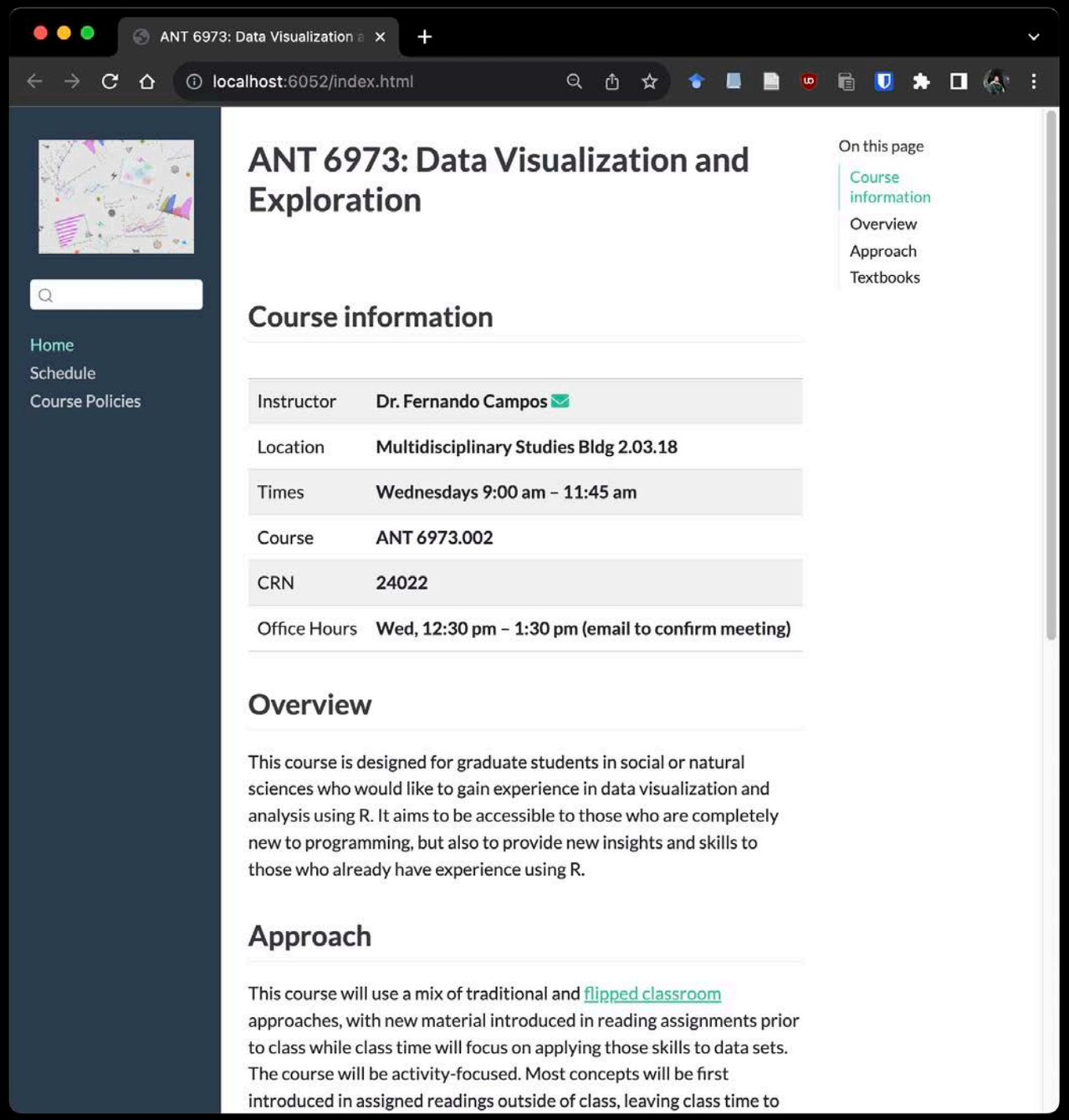
# ORIENTATION

# ORIENTATION

- Syllabus and course website
- R and RStudio

# COURSE WEBSITE

- https://www.campos-lab.net/courses/data-viz/
- Bookmark it!
- Go here for up-to-date information, schedule, assignments, readings, etc.
- I'll use Canvas only for posting grades.
- Let's explore this a bit…



# ORIENTATION

- Syllabus and course website
- R and RStudio

# WHAT'S R VS. RSTUDIO?

- R is a programming language.
- RStudio is a nice user interface for R (an *integrated development environment* or IDE) that provides many convenient features and tools for using R.



Engine

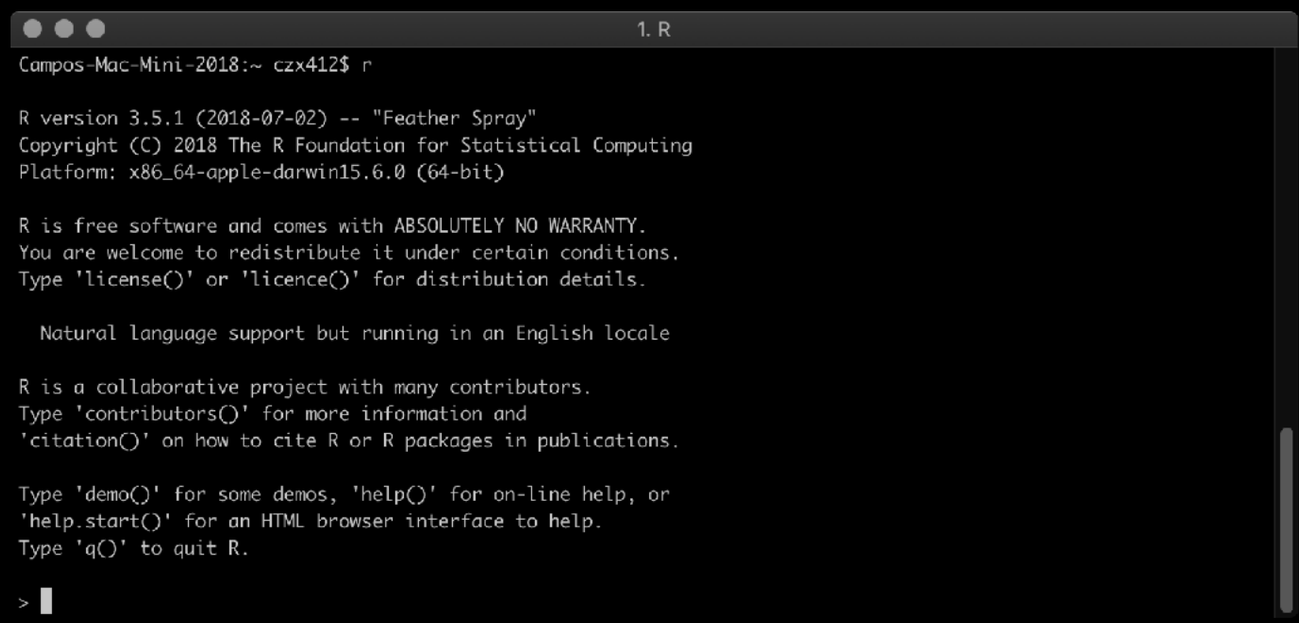


Dashboard

# WHY R?



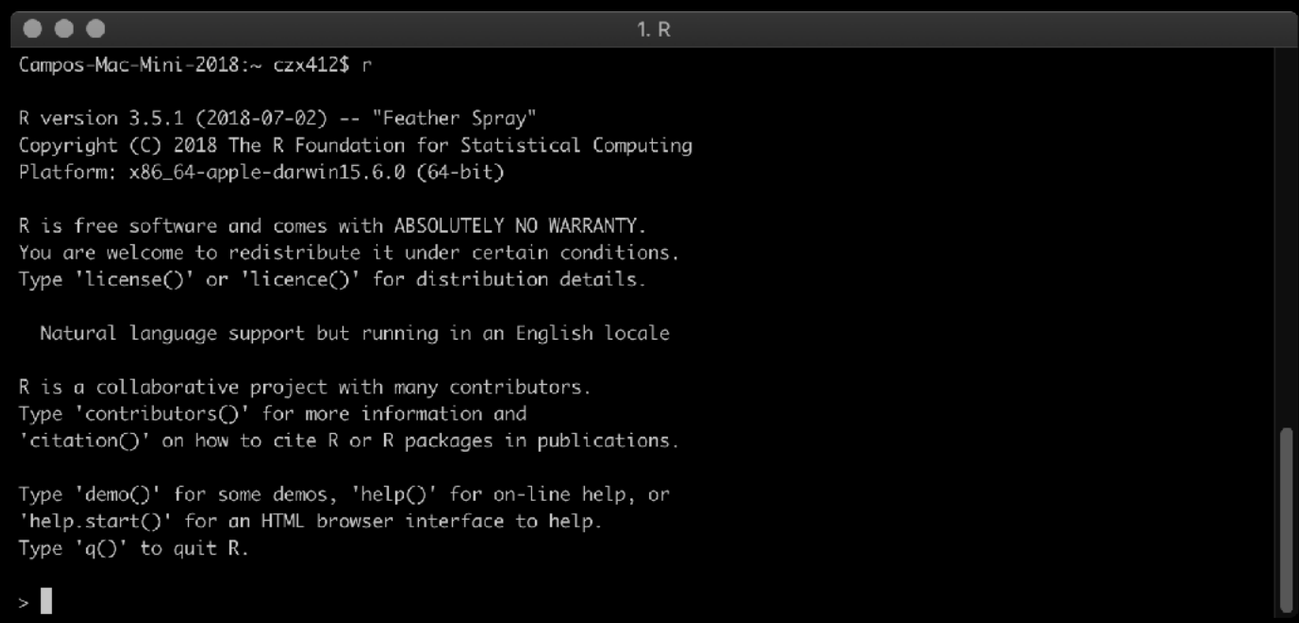
- R is the premier data analysis and visualization platform.



# WHY LEARN R?



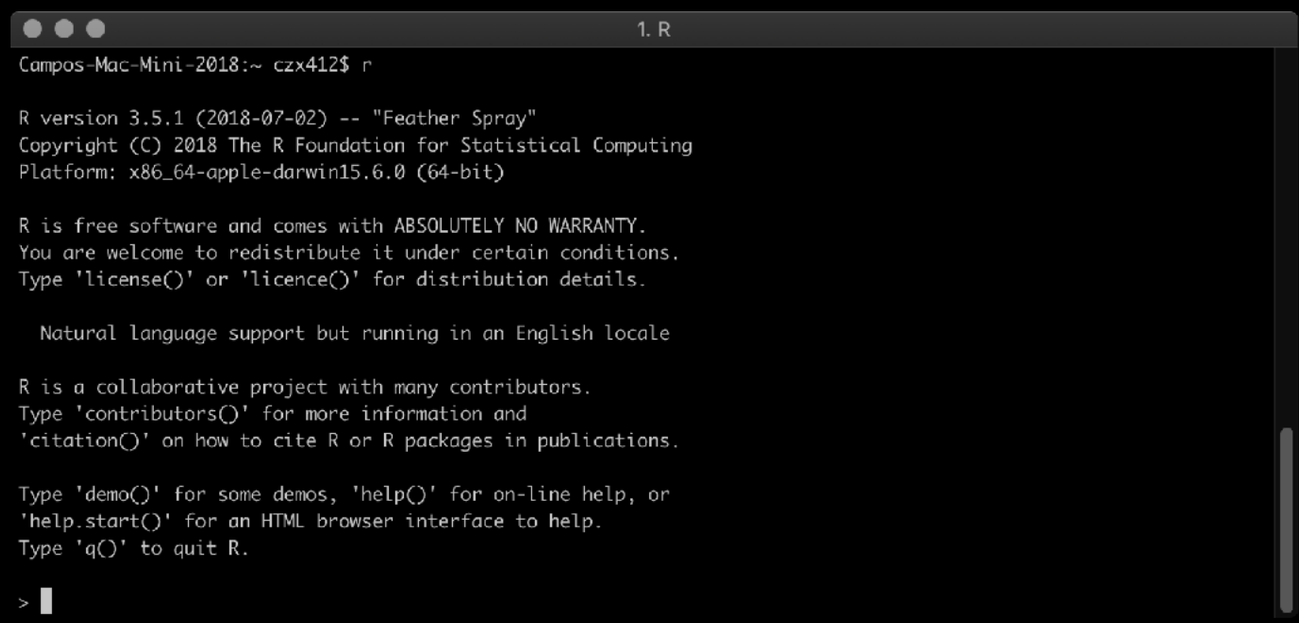
- **Pros (1 of 2):**
  - Powerful, state-of-the-art, and used by many researchers, R is becoming a standard for data analysis
  - Easy to extend, modify, and improve with add-on packages
    - Programmable: if R can't do a particular task right now, you (or someone) can program it
  - Freely available for Windows, Mac, Unix



# WHY LEARN R?

- **Pros (2 of 2):**
  - Reproducibility: R relies on a series of *written commands* rather than pointing and clicking (and this is a good thing!)
  - Powerful plotting packages produce publication-quality graphics.
  - R has a large and welcoming community.

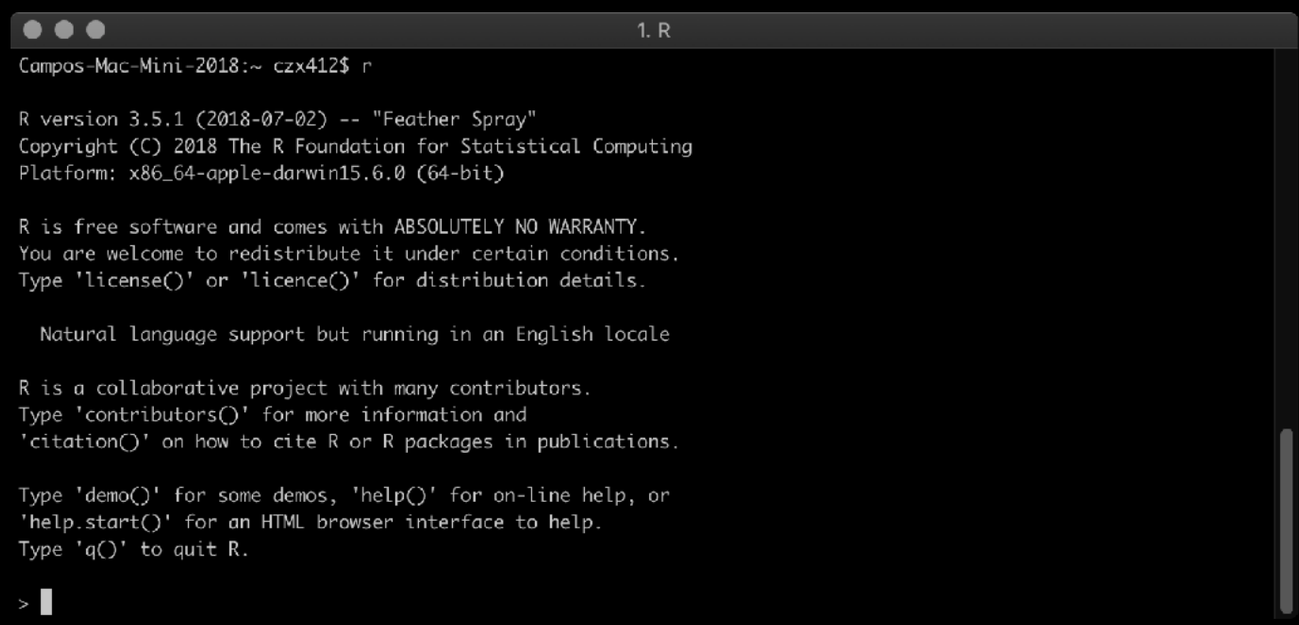




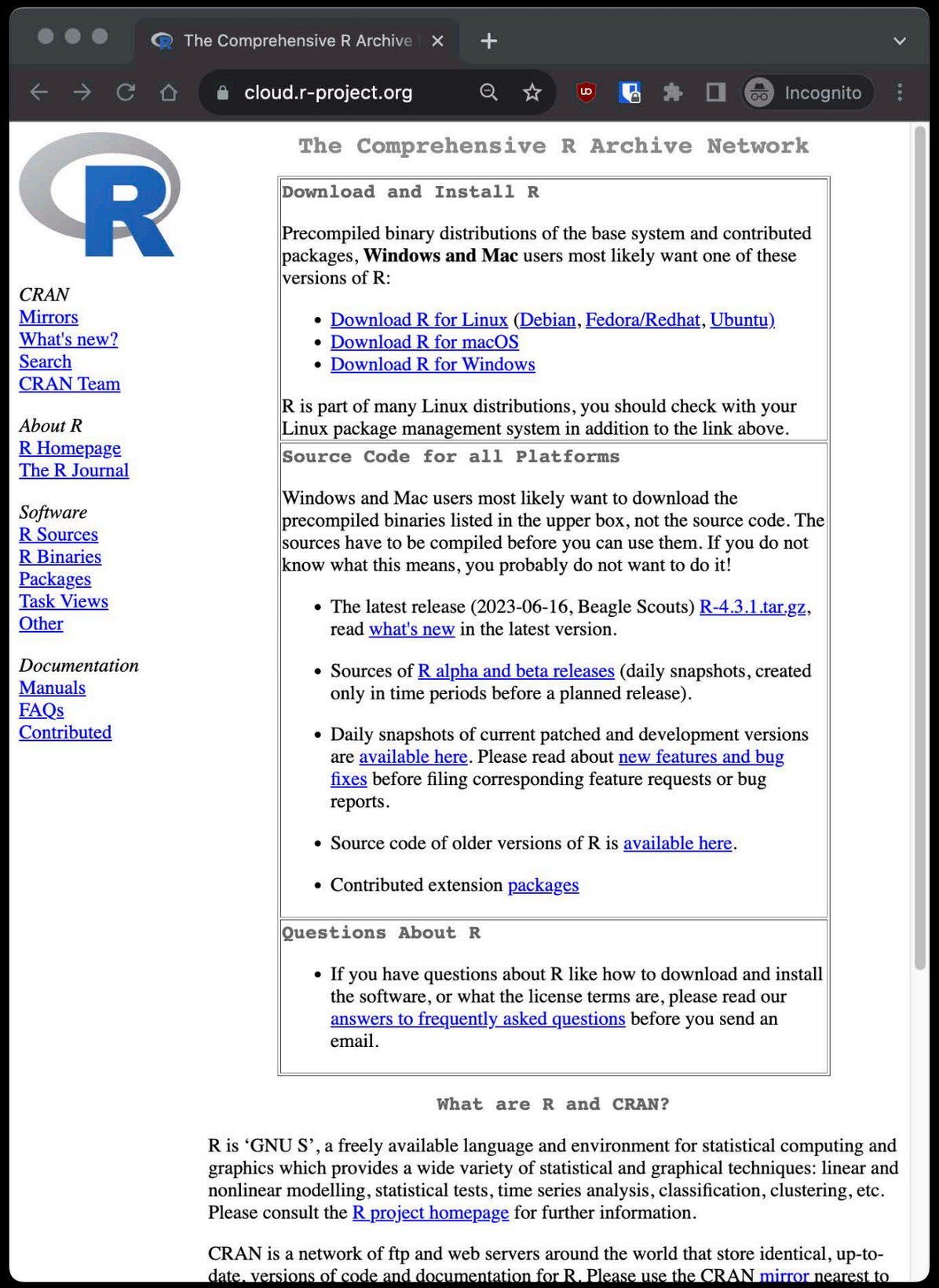
# WHY LEARN R?



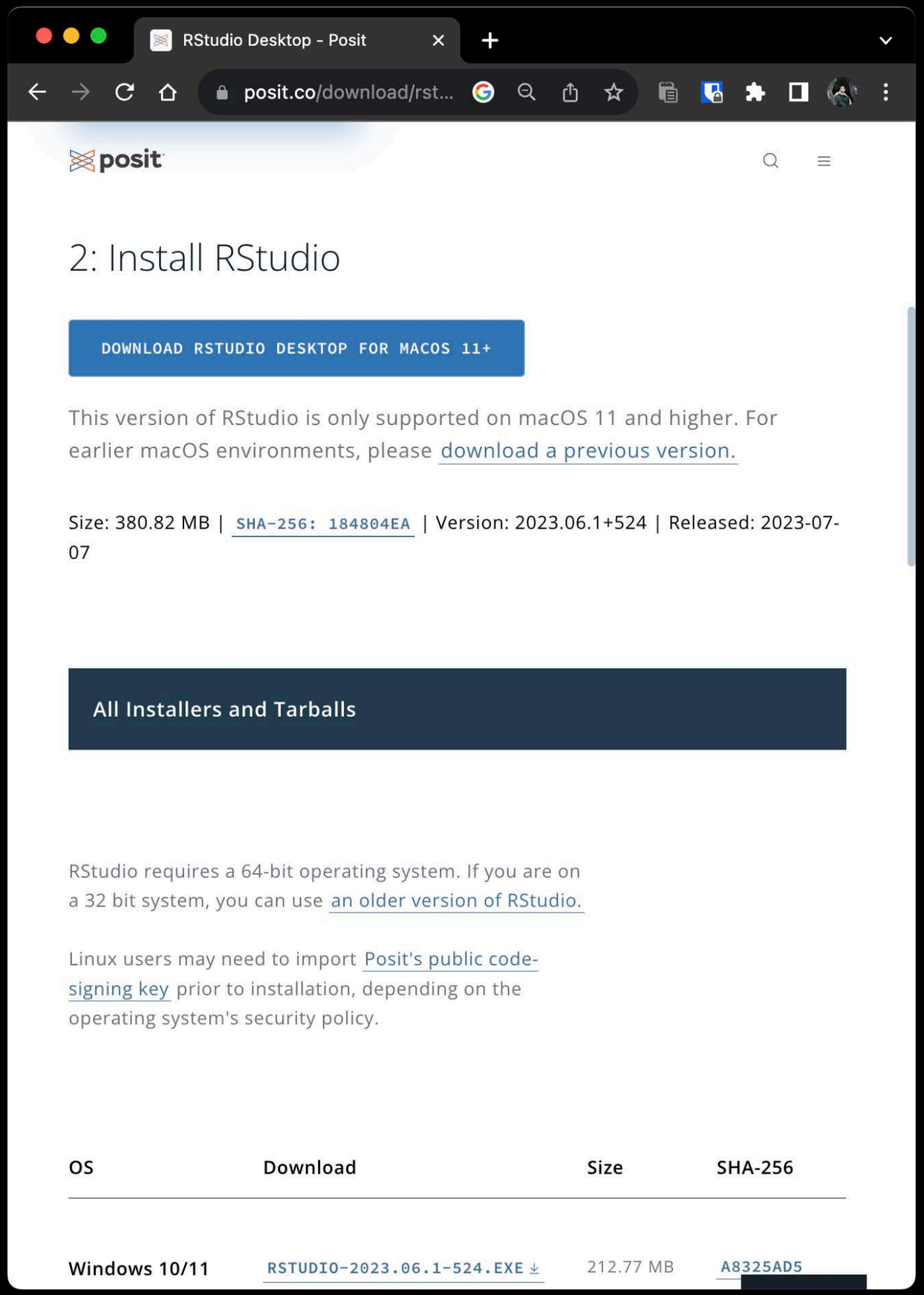
- **Cons:**
  - Command-based rather than interactive in the point-and-click sense (like SPSS & Excel)
    - Can be significant learning hurdle
    - Specific commands easy to forget
  - Documentation sometimes poor
  - Still evolving: backward-compatibility has been an issue



- Let’s make sure we all have a current installation of R.
- https://cloud.r-project.org/
- If you already have R installed, it's still a good idea to update.



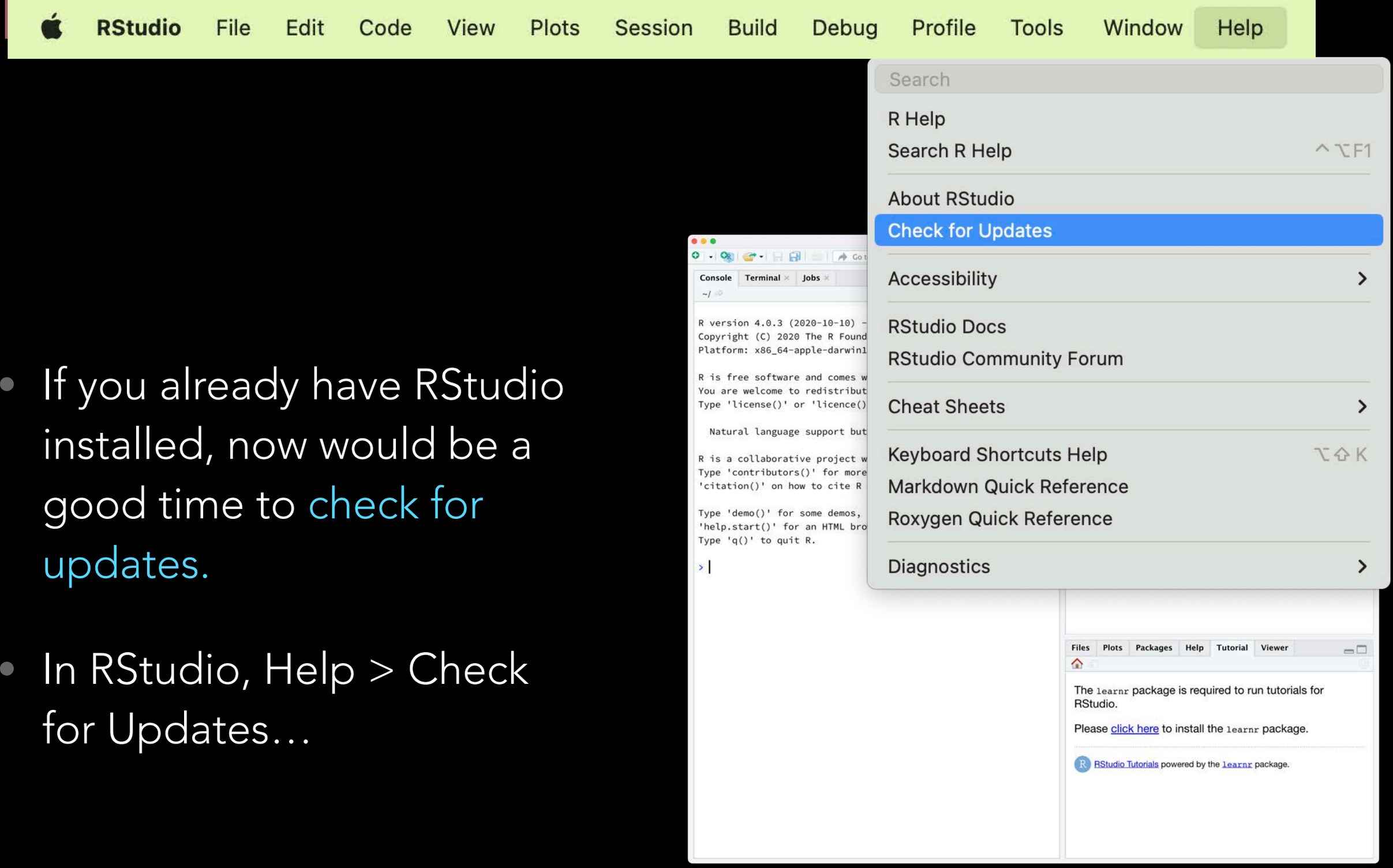
- And a current installation of RStudio.
- https://posit.co/download/rstudio-desktop/
- If you already have RStudio installed, check for a new version using menus:
  - Help > Check for Updates



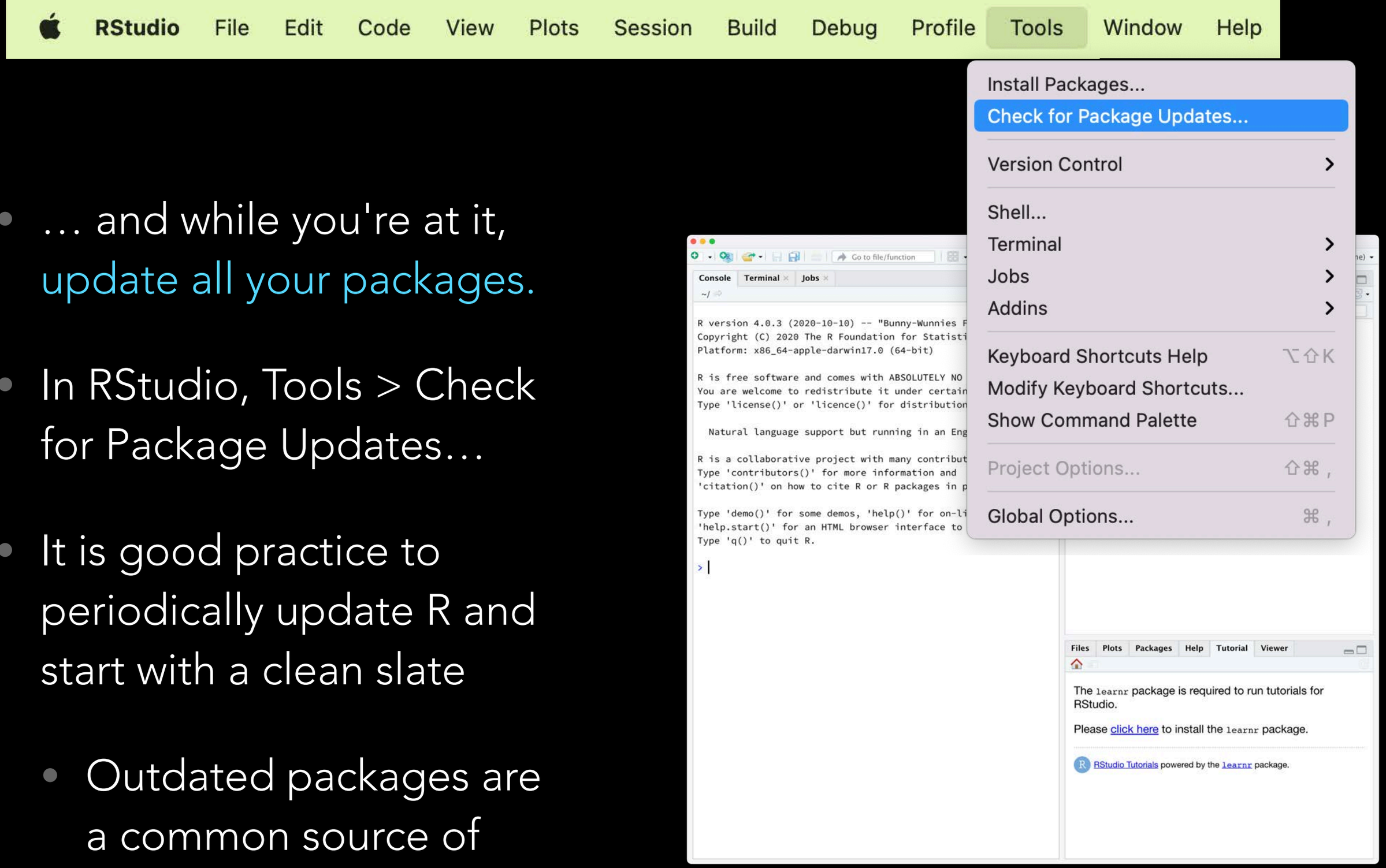
LET'S PAUSE UNTIL WE ALL COMPLETE THESE STEPS

- If you already have RStudio installed, now would be a good time to check for updates.

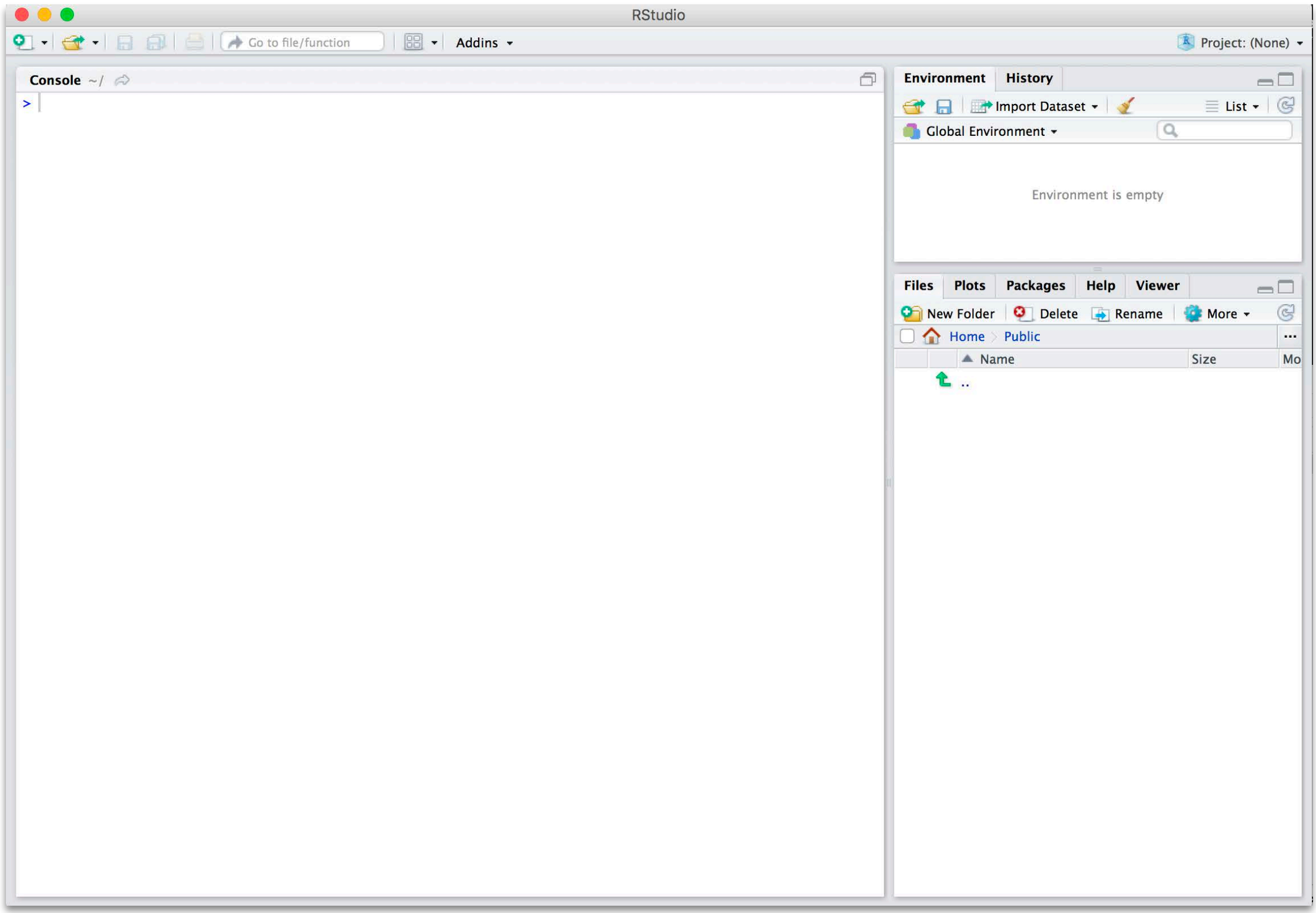
- In RStudio, Help > Check for Updates…



- … and while you're at it, update all your packages.
- In RStudio, Tools > Check for Package Updates…
- It is good practice to periodically update R and start with a clean slate
  - Outdated packages are a common source of errors




RStudio


RStudio
Go to file/function
Addins
Project: (None)
Console ~/
>
Environment
History
Import Dataset
List
Global Environment
Environment is empty
Files
Plots
Packages
Help
Viewer
New Folder
Delete
Rename
More
Home
Public
Name
Size
Mo
..

RStudio

Go to file/function | Addins | Project: (None)

Untitled1 | Source on Save | Run | Source

1

SOURCE

1:1 | (Top Level) | R Script

Environment | History | Connections

Import Dataset | List

Global Environment

ENVIRONMENT

Console | Terminal

~/

CONSOLE

Files | Plots | Packages | Help | Viewer

New Folder | Delete | Rename | More

Home > OneDrive > Temp

Name | Size | Modified

..

VIEWER

RStudio

Go to file/function | Addins | Project: (None)

Untitled1 | Source on Save | Run | Source

1

# SOURCE

Write and run code that you intend to keep.

1:1 | (Top Level) | R Script

Environment | History | Connections

Import Dataset | List

Global Environment

# ENVIRONMENT

Console | Terminal

~/

# CONSOLE

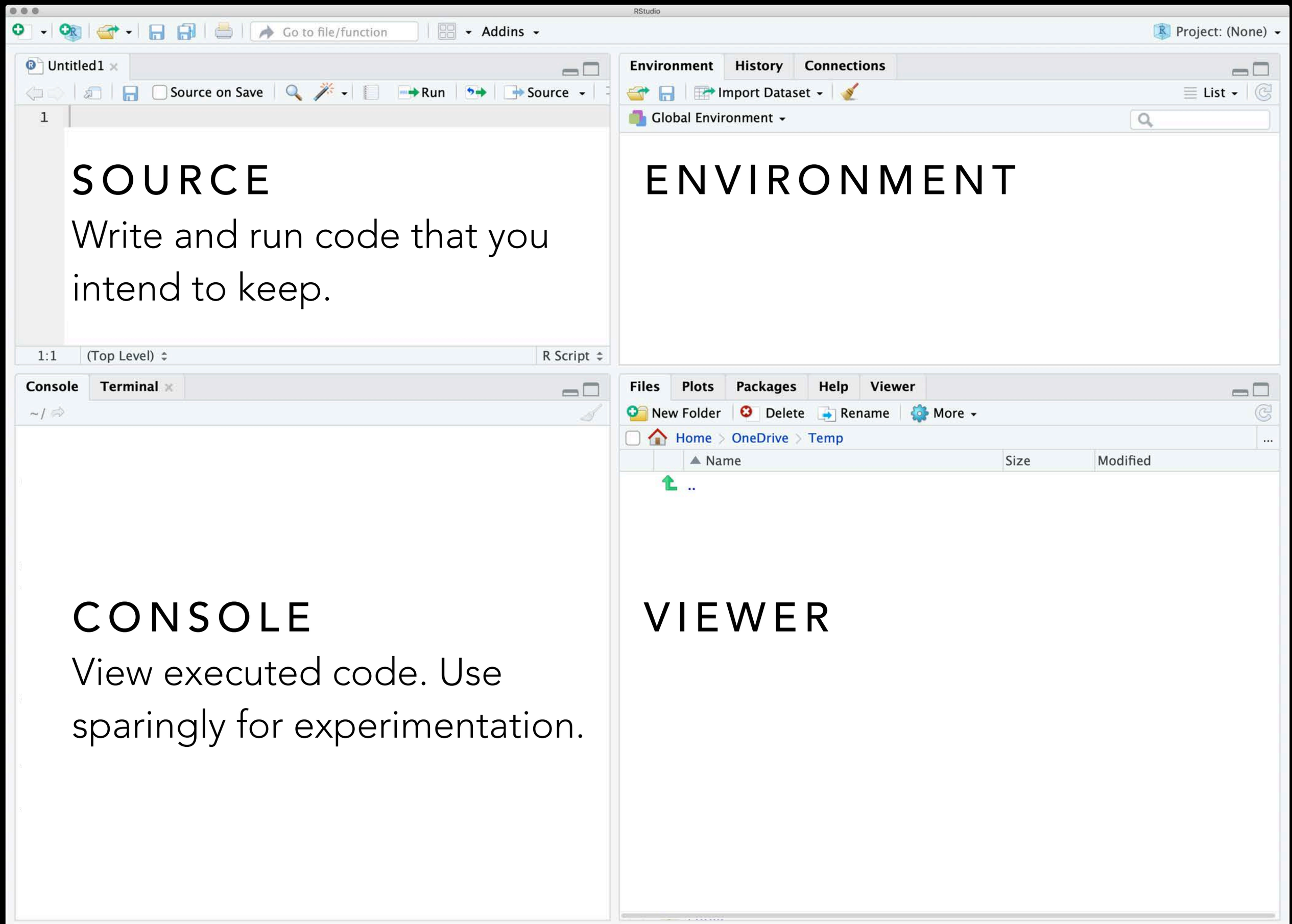
Files | Plots | Packages | Help | Viewer

New Folder | Delete | Rename | More

Home > OneDrive > Temp

| Name | Size | Modified |
|---|---|---|
| .. | | |

# VIEWER




## SOURCE

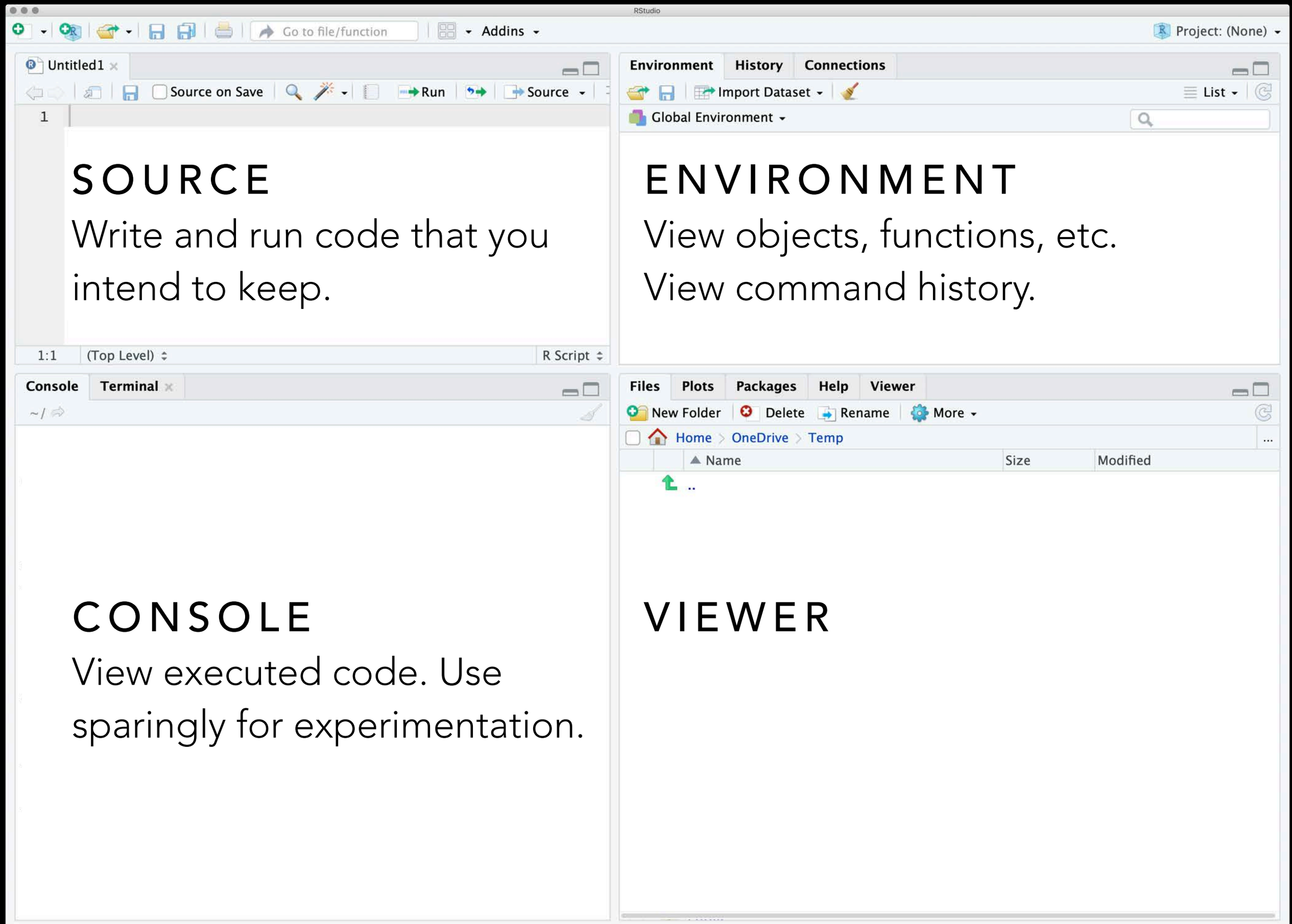
Write and run code that you intend to keep.

## ENVIRONMENT

## CONSOLE

View executed code. Use sparingly for experimentation.

## VIEWER




## SOURCE

Write and run code that you intend to keep.

## ENVIRONMENT

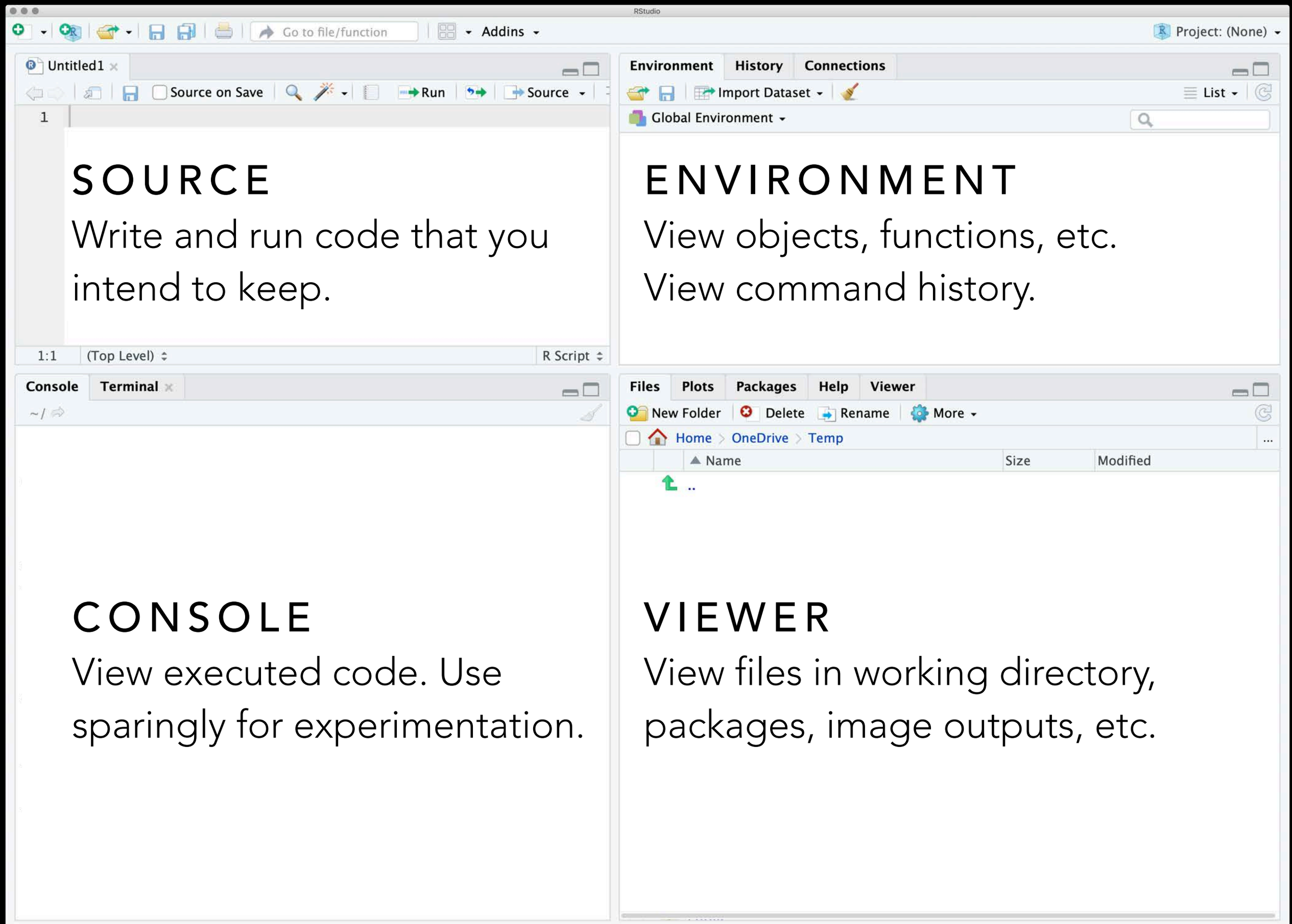
View objects, functions, etc.

View command history.

## CONSOLE

View executed code. Use sparingly for experimentation.

## VIEWER




## SOURCE

Write and run code that you intend to keep.

## ENVIRONMENT

View objects, functions, etc.

View command history.

## CONSOLE

View executed code. Use sparingly for experimentation.

## VIEWER

View files in working directory, packages, image outputs, etc.

WEEK1

# R BASICS

# TOPICS

- Some R basics
- Dealing with R Packages

# YOUR TURN

Please complete the activity for this week listed on the course website under:

Schedule > Assignments

It's short!






# BASIC R: IMPORTANT CONCEPTS

- "Running code" involves giving commands to R that are executed in the Console pane.

- You can type each line of code in the Console, but it is a better idea to write a *draft of your code as you go* by using an R script.

- A script is just a text file that you save R code in.

- Use Cmd + Enter (Mac) or Ctrl + Enter (Windows) to run selected lines from your R script.

# BASIC R: IMPORTANT CONCEPTS

- Comments (lines that R does not try to evaluate) start with #
    - Good code includes lots of comments!
- Computers are stupid: R can't guess what you want to do. Attention to detail is important!

# BASIC R: IMPORTANT CONCEPTS

- Values are saved in *objects*, which have names.
  - Names are case-sensitive!
  - When creating objects, aim for short, meaningful names.
    - Style guides recommend names in all lowercase, with words separated by underscores (e.g., `my_name`).
    - Valid names in R cannot start with a number or include the following special symbols: `^ ! $ @ + - / * [space]`
- Create new objects using the assignment operator (`<-` which will appear as ← hereafter)

# BASIC R: IMPORTANT CONCEPTS

- You do things using operators and functions.
  - Example operators: `+`, `←`
  - Example functions: `c()`, `mean()`, `summary()`

```
# Combine multiple elements into a vector with c()
primes ← c(2, 3, 5, 7, 11, 13)
```

# BASIC R: OPERATORS

## ASSIGNMENT OPERATORS

| OPERATOR | DESCRIPTION |
|---|---|
| ← | ASSIGN A VALUE TO A NAME |
| = | SAME BUT MORE LIMITED (DON'T USE FOR NOW) |

```
my_name ← "Fernando Campos"
my_height ← 182
```

Tip: use RStudio shortcut (`⌥` + `-` in macOS, `Alt` + `-` in Windows)

# BASIC R: OPERATORS

## ARITHMETIC OPERATORS

| OPERATOR | DESCRIPTION |
|---|---|
| + | ADDITION |
| - | SUBTRACTION |
| * | MULTIPLICATION |
| / | DIVISION |
| ^ | EXPONENTIATION |

```
2 + 4
2 - 4
2 * 4
2 / 4
2 ^ 4
```

# BASIC R: OPERATORS

## COMPARISON OPERATORS

| OPERATOR | DESCRIPTION |
|---|---|
| < | LESS THAN |
| <= | LESS THAN OR EQUAL TO |
| > | GREATER THAN |
| >= | GREATER THAN OR EQUAL TO |
| == | EXACTLY EQUAL TO |
| != | NOT EQUAL TO |

# BASIC R: OPERATORS

## COMPARISON OPERATORS

| OPERATOR | DESCRIPTION |
|---|---|
| < | LESS THAN |
| ⩽ | LESS THAN OR EQUAL TO |
| > | GREATER THAN |
| ⩾ | GREATER THAN OR EQUAL TO |
| ⩵ | EXACTLY EQUAL TO |
| ≠ | NOT EQUAL TO |

Or using font with ligatures:
(Jetbrains mono)



# BASIC R: OPERATORS

## COMPARISON OPERATORS

| OPERATOR | DESCRIPTION |
|---|---|
| < | LESS THAN |
| ⩽ | LESS THAN OR EQUAL TO |
| > | GREATER THAN |
| ⩾ | GREATER THAN OR EQUAL TO |
| == | EXACTLY EQUAL TO |
| ≠ | NOT EQUAL TO |

```
# What kind of values will be returned?
2 < 4
2 ⩾ 4
2 == 4
2 ≠ 4
```

# BASIC R: OPERATORS

## LOGICAL OPERATORS

| OPERATOR | DESCRIPTION |
|---|---|
| ! | NOT |
| \| | OR |
| & | AND |

```
!TRUE
TRUE | FALSE
TRUE & FALSE
(1 < 5) & (7 > 1)

my_name ← "Fernando Campos"
my_height ← 182

(my_height > 190) | (my_name == "Fernando Campos")
```

# BASIC R: IMPORTANT CONCEPTS

- R comes with many built-in functions, and packages provide additional specialized functions.
- Data that you pass into a function is called the function's argument.
- Some arguments are optional, some are required. Optional arguments have a default value if none is specified.
- Arguments have names, which you *can* (but usually should not) omit.
- Nested functions are evaluated inside to outside.

```
round(3.1415)
round(x = 3.1415, digits = 2)
round(mean(1:6), digits = 0)
```

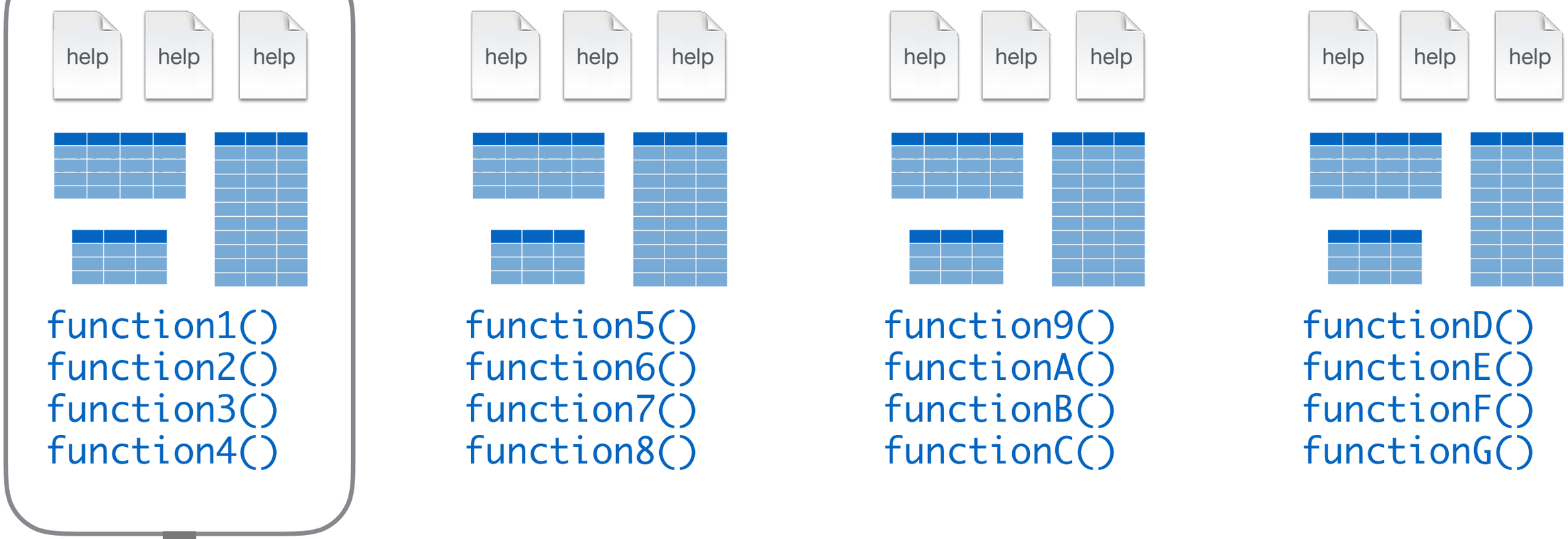
# BASIC R: IMPORTANT CONCEPTS

- R uses *element-wise execution*
    - An operation on an object with multiple element is applied to each element in the set.
    - When both objects have multiple elements, they elements are matched up by position and the operations are applied separately to each pair.

```
dice ← 1:6
dice * 10
dice * dice
dice + c(10, 100)
```

# R PACKAGES

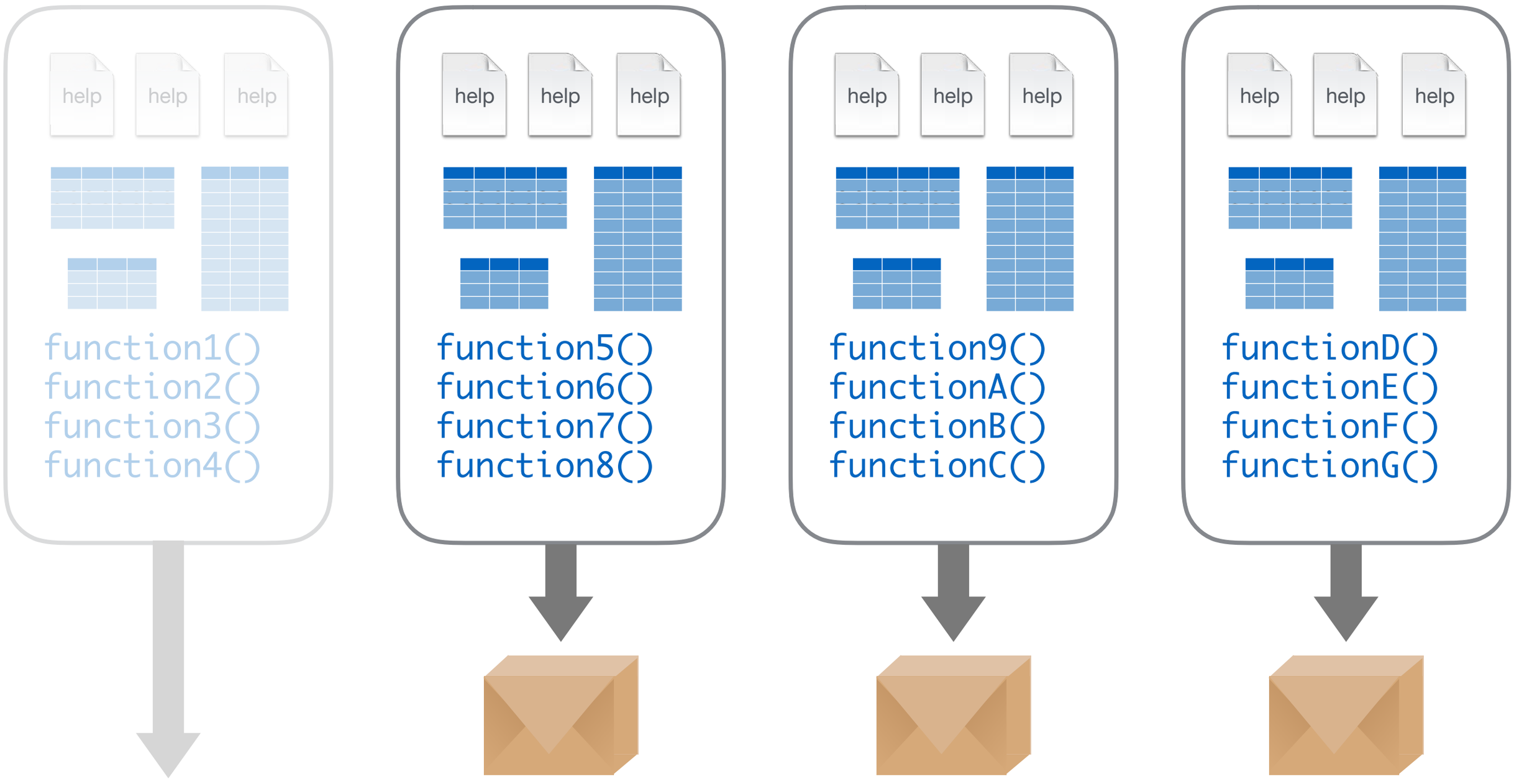
# R PACKAGES




Base R

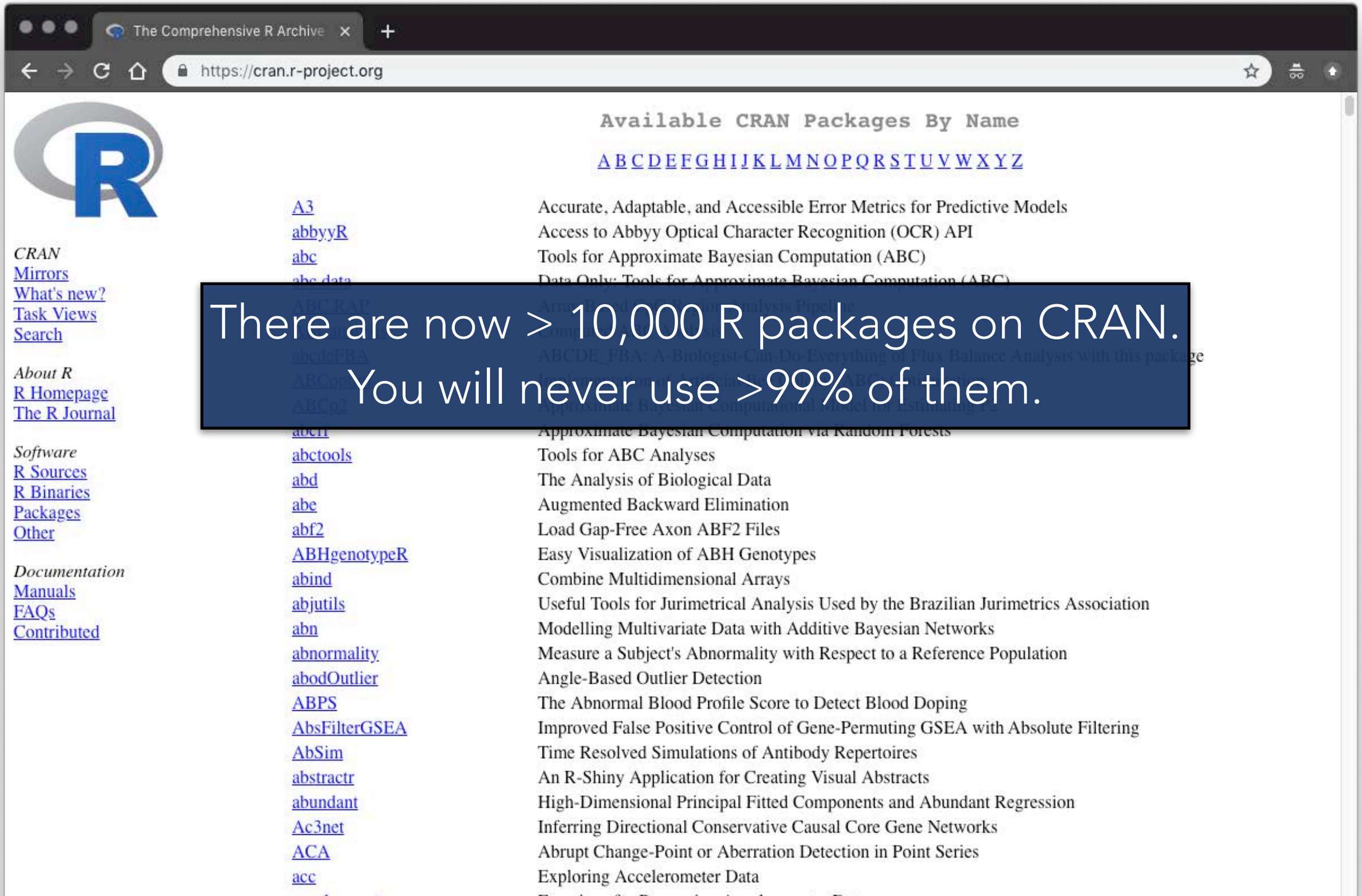
CC by RStudio. https://github.com/rstudio/master-the-tidyverse

# R PACKAGES




CC by RStudio. https://github.com/rstudio/master-the-tidyverse

# R PACKAGES




There are now > 10,000 R packages on CRAN.
You will never use >99% of them.

# USING PACKAGES

## 1

```
install.packages("foo")
```

Downloads files to computer

**1 x per computer**

## 2

```
library("foo")
```

Loads package

**1 x per R Session**

CC by RStudio. https://github.com/rstudio/master-the-tidyverse


RStudio®

# 1. ORGANIZE YOUR WORK INTO PROJECTS

(more on this next week)




# 2. ALWAYS START A SESSION WITH A CLEAN SLATE

RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help

Install Packages...
Check for Package Updates...
Version Control
Shell...
Terminal
Jobs
Addins
Keyboard Shortcuts Help ⌥⇧K
Modify Keyboard Shortcuts...
Show Command Palette ⇧⌘P
Project Options... ⇧⌘,
Global Options... ⌘,

RStudio

Go to file/function | Addins

Console | Terminal | Jobs

~/Documents/academic-website/

```
R version 4.0.3 (2020-10-10) -- "Bunny-Wunnies Freak Out"
Copyright (C) 2020 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>
```

Environment | History | Connections

Import Dataset | R | Global Environment

Files | Plots | Packages | Help | Tutorial | Viewer

New Folder | Delete | Rename | More

Home > Documents > academic-website

| Name | Size | Modified |
|---|---|---|
| .. | | |
| .gitignore | 53 B | Apr 12, 2020, 9:53 AM |
| .Rhistory | 2 KB | Jan 14, 2021, 10:44 AM |
| academic.Rproj | 258 B | Jan 14, 2021, 10:44 AM |
| assets | | |
| config | | |
| content | | |
| data | | |
| LICENSE.md | 1.1 KB | Apr 12, 2020, 9:51 AM |
| public | | |
| README.md | 3.3 KB | Apr 12, 2020, 9:51 AM |
| resources | | |

RStudio

Go to file/function | Addins | Project: (None)

Console | Terminal | Jobs

~/Documents/academic-website/

```
R version 4.0.3 (2020-10-10) -- "Bunn
Copyright (C) 2020 The R Foundation f
Platform: x86_64-apple-darwin17.0 (64

R is free software and comes with ABS
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Type 'license()' or 'licence()' for d

  Natural language support but runnin

R is a collaborative project with man
Type 'contributors()' for more inform
'citation()' on how to cite R or R pa

Type 'demo()' for some demos, 'help()
'help.start()' for an HTML browser in
Type 'q()' to quit R.

>
```

Options

- General
- Code
- Console
- Appearance
- Pane Layout
- Packages
- R Markdown
- Sweave
- Spelling
- Git/SVN
- Publishing
- Terminal
- Accessibility
- Python

Basic | Graphics | Advanced

**R Sessions**

Default working directory (when not in a project):

~/Documents/academic-website | Browse...

- [x] Restore most recently opened project at startup
- [x] Restore previously open source documents at startup

**Workspace**

- [ ] Restore .RData into workspace at startup

Save workspace to .RData on exit: Never

**History**

- [x] Always save history (even when not saving .RData)
- [x] Remove duplicate entries in history

**Other**

- [ ] Wrap around when navigating to previous/next tab
- [x] Automatically notify me of updates to RStudio
- [ ] Send automated crash reports to RStudio

OK | Cancel | Apply

List

t is empty

iewer

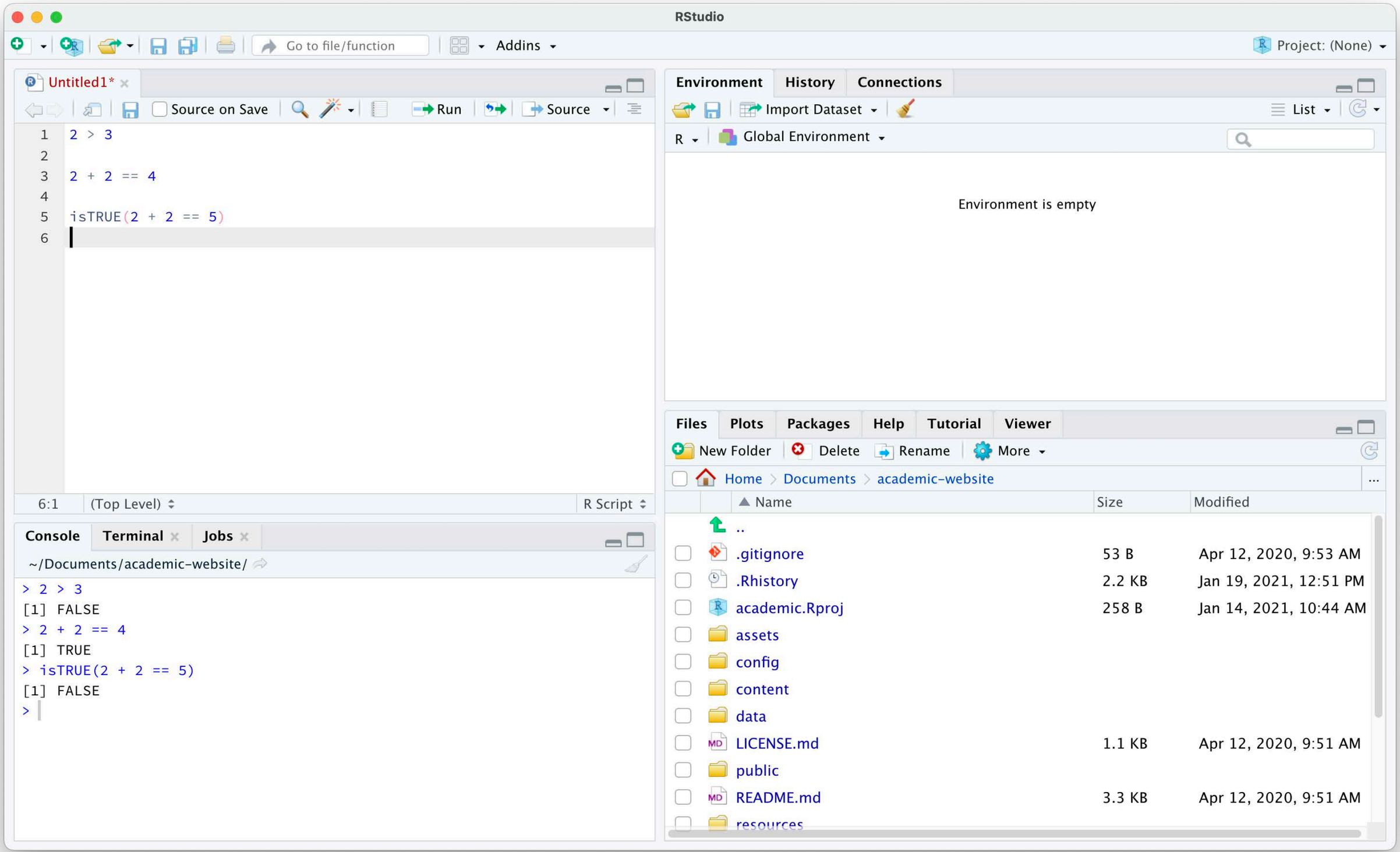
re

...

| | Size | Modified |
|---|---|---|
| | 53 B | Apr 12, 2020, 9:53 AM |
| | 2 KB | Jan 14, 2021, 10:44 AM |
| | 258 B | Jan 14, 2021, 10:44 AM |
| | 1.1 KB | Apr 12, 2020, 9:51 AM |
| | 3.3 KB | Apr 12, 2020, 9:51 AM |
| resources | | |

# 3. TWO WAYS OF INTERACTING WITH R: CONSOLE AND SCRIPTS

(most of the time, use scripts)


RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
Source on Save
Run
Source
2 > 3
2 + 2 == 4
isTRUE(2 + 2 == 5)
6:1
(Top Level)
R Script
Console
Terminal
Jobs
~/Documents/academic-website/
> 2 > 3
[1] FALSE
> 2 + 2 == 4
[1] TRUE
> isTRUE(2 + 2 == 5)
[1] FALSE
>
Environment
History
Connections
Import Dataset
List
R
Global Environment
Environment is empty
Files
Plots
Packages
Help
Tutorial
Viewer
New Folder
Delete
Rename
More
Home
Documents
academic-website
Name
Size
Modified
..
.gitignore
53 B
Apr 12, 2020, 9:53 AM
.Rhistory
2.2 KB
Jan 19, 2021, 12:51 PM
academic.Rproj
258 B
Jan 14, 2021, 10:44 AM
assets
config
content
data
LICENSE.md
1.1 KB
Apr 12, 2020, 9:51 AM
public
README.md
3.3 KB
Apr 12, 2020, 9:51 AM
resources




# 4. BECOME COMFORTABLE WITH ERROR MESSAGES




```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> my_data <- read.csv("my_data.csv")
Error in file(file, "rt") : cannot open the connection
In addition: Warning message:
In file(file, "rt") :
  cannot open file 'my_data.csv': No such file or directory
> Can you say that in plain English please?
Error: unexpected symbol in "Can you"
> :(
Error: unexpected ':' in ":"
>
```




# 5. USE RSTUDIO'S TOOLS TO MAKE LIFE EASIER